

STATEMENT OF PURPOSE – AMAZON ML SUMMER SCHOOL

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Before even getting into college, I had always heard about AI and ML here and there, so naturally, it caught my attention. I searched about it and got a basic idea of what AI and ML meant, but one question stayed with me: how? The concept didn't come to me intuitively at first. I had to watch videos, read articles, search endlessly, and even ask AI about AI before I slowly started understanding what was going on beneath all the hype.

My first proper exposure to these concepts came through a 14-day Python and AI/ML workshop organised in my college during my third semester. I learnt about them in much greater depth, but it also made me aware of just how deep the field is. I realised that being surrounded by terms like "AI" and "machine learning" was very different from actually understanding and applying them. I participated in a few hackathons where, with the help of my teammates and my own curiosity, I gained more practical exposure. In one of them, we built a model to analyse multilingual sentiment on Twitter. Since India has many regional languages, we first translated the text into English using the Google Translate API. After cleaning and preprocessing the data, we performed sentiment analysis using Hugging Face's RoBERTa model along with predefined keywords. Working on this project taught me that despite having pre-trained models, it is still difficult to always fit them into our projects. For example, our model was missing the intelligence required to tell apart sarcastic comments from genuine ones. It was still a better opportunity than copying a project from tutorials.

Despite all these experiences, which have strengthened my interest in the field, I still hesitate to confidently say, "Yes, I know ML." I have enough exposure to appreciate what the field can do, but I also know how much I still don't understand. Rather than letting it discourage me, I take it as motivation. I want to understand how these ML tools actually work, not just use existing models. I believe the Amazon ML Summer School will help me bridge that gap. Learning from and interacting with scientists at Amazon who have extensive experience in the field will provide me with the structured guidance I've been seeking. More importantly, this intensive course will help me deepen my understanding of machine learning. After the program, I hope to be able to apply my learnings and create the impressive projects I've seen around me on my own as well, and turn this into the field I pursue in the future.